SafePath

Survival tools that work when the network does not.

Offline-first disaster preparedness, family safety,
community coordination & AI-assisted guidance
for Sri Lanka

Professional Pitch Document
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Prepared for judges, investors, government partners,
NGOs, and emergency response stakeholders

Team Hades · SafePath
github.com/godragun/SafePath

Evidence standard: Features described as shipped are implemented in the SafePath Android codebase (Kotlin, Room, Hilt, Supabase, Gemini, Nearby Connections). Items labeled **Future proposal** are not claimed as built.

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01 Executive Summary

SafePath is an offline-first Android application designed for disaster preparedness and response in Sri Lanka. When mobile networks fail—as they often do during floods, landslides, and storms—citizens still need checklists, survival guidance, shelter awareness, and a way to tell loved ones they are safe. SafePath keeps those capabilities on-device, then adds cloud verification, Family Circle, community chat, AI assistance, and peer-to-peer mesh notes when connectivity returns.

Why it matters. Sri Lanka faces recurring monsoon floods and landslides. During the May–June 2024 severe weather period, national relief reporting indicated more than 235,000 people affected, dozens of fatalities, and thousands of damaged homes (NDRSC / ECHO situation reporting via ReliefWeb). In such events, delayed information, fragmented family communication, and dependence on always-online apps compound risk—especially in districts such as Ratnapura, Kalutara, and Galle that SafePath already seeds with verified shelter data.

Who suffers. Households in hazard-prone districts, elderly and mobility-limited residents, families separated during evacuation, tourists unfamiliar with local shelters, schools and clinics coordinating under stress, and responders who lack ground-truth from communities when towers are down.

Why existing tools fall short. Many consumer apps assume continuous data. Official channels are authoritative but not always interactive or personalized. Generic chat apps are not disaster-workflow tools. SafePath combines offline survival content, verified family safety broadcasts, community hazard notes that can hop device-to-device, and Gemini-powered guidance that can be cached for reuse offline.

Value in one line: SafePath turns a phone into a resilient personal emergency kit—checklists, guidance, shelters, family ‘I’m Safe’, community mesh notes, and AI Q&A—designed for Sri Lanka’s disaster reality.

02 The Problem

Disaster management fails people not only because hazards are severe, but because **information, coordination, and trust break at the worst moment**.

Core failure modes observed in real emergencies:

Problem	Why it hurts
Delayed response	Citizens wait for confirmation while roads flood; minutes matter for landslide and flash-flood risk.
Unreliable information	Rumors spread on social media; offline households miss updates entirely.
Communication collapse	Congested or damaged networks block calls and cloud-only apps.
Family separation	Parents, elders, and children evacuate separately with no shared safety signal.
Poor resource awareness	People do not know which shelters are nearby or whether they are verified.
Community coordination gaps	Local hazard sightings (blocked roads, needs, ‘area safe’) never reach neighbors without data.
Preparedness debt	Households lack personalized go-bags and actionable checklists before the season peaks.

Sourced context (not a SafePath metric): NDRSC reporting for 15 May–11 June 2024 described >235,000 people affected by floods/landslides in Sri Lanka, with 37 deaths and >16,000 houses damaged (ECHO Daily Flash / ReliefWeb). Ratnapura and Kalutara were among the hardest-hit districts—exactly the kind of geography SafePath seeds with shelter records today.

03 Our Solution

SafePath is a **layered resilience system**: offline core first, cloud and AI when available, peer mesh when neither works well.

End-to-end workflow (as implemented)

Stage	What the user does	What the system does
1. Onboard	Sign up / sign in, or continue as guest	Encrypted session storage; optional email confirmation deep link
2. Verify	Phone or email OTP	Unlocks Family Circle via users.verified on Supabase + Room cache
3. Prepare	Emergency pack + optional AI personalization	Checklist in Room; Gemini pack saved for offline reuse
4. Learn	Survival library / Ask SafePath AI	Offline guides; Q&A cached (SHA-256 key, 30-day TTL)
5. Monitor	Home dashboard	Alert status, Family Circle entry, map CTA, Sri Lanka silhouette + shelter dots
6. Report / navigate	FAB report alert; open evacuation map	Supabase alerts + Leaflet map with shelters/GPS; Google Maps handoff
7. Coordinate	I'm Safe; chat; Mesh Notes	WorkManager queues; Nearby P2P hops; cloud sync when online
8. Escalate	Imminent alert screen	Guided 'leave now' path toward the map

Decision logic (product)

- **Connectivity-aware:** NetworkMonitor gates online chat/sync vs mesh/queue paths.
- **Verification-gated trust:** Family Circle requires verified identity—not anonymous spam contacts.
- **Cache-first AI:** Ask once online; reuse offline when the same question is asked again.
- **Local-first mesh:** Notes always persist in Room before hopping or syncing.

Architecture snapshot: UI (Fragments) → ViewModels → Repositories → Room + WorkManager + Nearby Connections | Supabase (Auth, Postgrest, Realtime, Storage) + Gemini 2.5 Flash.

04 Feature Analysis

Each feature below is implemented in code. Impact is described in operational terms—not marketing slogans.

Authentication & OTP verification

What it does: Email/password auth, check-email flow, deep-link session restore, phone or email OTP via Supabase, encrypted session prefs.

Why it matters in emergencies: Establishes a trusted identity before family safety features unlock—reducing abuse of 'I'm Safe' and contact lists.

Technical anchor: *auth/, VerificationRepository, SessionSecureStorage, PhoneVerificationFragment*

Family Circle + I'm Safe

What it does: Trusted contacts (free tier limit = 1); GPS (or last-known/fallback) safety broadcast to Supabase; offline queue + SafeBroadcastWorker.

Why it matters in emergencies: Gives families a single intentional safety signal when calls fail—queued until the network returns.

Technical anchor: *FamilyCircleRepository, SafeBroadcastRepository, FamilyCircleLimits*

Emergency pack checklists

What it does: Tap-to-check items with progress; Room persistence in the cache database.

Why it matters in emergencies: Turns vague 'be prepared' advice into a measurable household readiness task.

Technical anchor: *EmergencyPackFragment, PackRepository, AppDatabase pack_items*

AI pack personalization (Gemini)

What it does: Household profile (size, kids, elderly, pets, medical needs) → Gemini JSON checklist + tips; stored offline; defaults if API fails.

Why it matters in emergencies: Adapts preparedness to real household constraints instead of one-size-fits-all lists.

Technical anchor: *GeminiPrepService, PersonalizedPackRepository*

Ask SafePath AI

What it does: Gemini Q&A for flood/landslide/cyclone/tsunami/pack/first-aid topics; Room cache with TTL.

Why it matters in emergencies: Provides calm, practical steps when users cannot reach a call center—and remembers answers for offline reuse.

Technical anchor: *GeminiQaService, AiQaRepository, AiQaCacheEntity*

Home dashboard & silhouette map

What it does: Alert cards, Family Circle, AI entry, map CTA; detailed Sri Lanka district SVG with dark shelter markers.

Why it matters in emergencies: Gives an at-a-glance safety posture and shelter awareness without opening a full map.

Technical anchor: *HomeFragment, map_silhouette.html, ShelterRepository*

Evacuation map

What it does: Leaflet WebView with user location, Room shelters, active alerts; navigate via Google Maps intent. *Basemap tiles currently require internet.*

Why it matters in emergencies: Helps users orient to shelters and reported hazards; true offline tiles are planned, not shipped.

Technical anchor: *EvacuationMapFragment, assets/map.html*

Report emergency alert

What it does: Bottom sheet: disaster type, severity, district, description, GPS → Supabase alerts.

Why it matters in emergencies: Turns citizen observations into structured hazard records responders and map users can see.

Technical anchor: *ReportAlertBottomSheet, AlertRepository*

Imminent alert flow

What it does: Dedicated screen with urgency UI; Home can navigate when severity is IMMINENT.

Why it matters in emergencies: Shortens the path from 'this is critical' to 'open the map / leave'.

Technical anchor: *ImminentAlertFragment*

Survival library

What it does: Offline step-by-step guides from local data/assets.

Why it matters in emergencies: Keeps life-saving instructions available with zero connectivity.

Technical anchor: *SurvivalLibraryFragment, SurvivalRepository*

Community online chat

What it does: Supabase realtime chat rooms with optional image storage.

Why it matters in emergencies: Supports neighborhood coordination when the network is up.

Technical anchor: *ChatRepository, ChatRoomFragment*

Mesh Notes (P2P)

What it does: Hazard/need/safe/info notes ≤280 chars; Room-first; Nearby Connections cluster; max 5 hops; 48h TTL; UUID dedupe; MeshSyncWorker to cloud.

Why it matters in emergencies: Creates a last-mile information fabric when cellular data is unavailable—critical for blocked roads and local needs.

Technical anchor: *MeshNotesRepository, P2pManager, MeshNotePayload*

Settings & dark UI

What it does: Material 3 theme, dark mode preference, notification preference flag.

Why it matters in emergencies: Improves usability in low-light / night evacuations; notification channel exists (delivery pipeline still preference-level).

Technical anchor: *SettingsFragment, Theme.SafePath*

Honest partials: On-device Dijkstra routing / DecisionEngine exist in code and tests but are **not wired into the evacuation map UI** (map uses Google Maps handoff). Offline OsmDroid tile archives are planned. Push notification posting is not fully productized beyond channel + preference. Sinhala/Tamil guidance columns exist but are placeholders.

05 Technical Innovation

SafePath's technical design prioritizes **operability under failure**: local persistence, queued sync, and peer transport—not cloud exclusivity.

Stack & architecture

Layer	Implementation
Client	Kotlin Android (API 26+), Material 3, Navigation, ViewBinding
DI	Hilt modules (App, Supabase, Gemini, Location)
Local data	Dual Room DBs: safepath_db (v8) + safepath_cache_db (v3)
Background	WorkManager + HiltWorkerFactory (safe broadcast + mesh sync)
Backend	Supabase Auth, Postgrest, Realtime, Storage
AI	Google Gemini 2.5 Flash via Retrofit generateContent
P2P	Google Nearby Connections (P2P_CLUSTER)
Maps UI	Leaflet WebView + SVG silhouette; OsmDroid dependency reserved
Location	Fused Location + LastLocationCache fallback
Security	BuildConfig secrets; EncryptedSharedPreferences sessions

Algorithms & AI (what exists)

- **Dijkstra routing engine** with hazardous-edge skipping, Haversine nearest-shelter selection, and point-in-polygon hazard checks — implemented under routing/; not yet the primary map UX.
- **Mesh epidemic-style forwarding:** hop count, TTL expiry, UUID deduplication.
- **Channel priority model:** cell-broadcast concept > mobile data > peer-to-peer (ChannelManager).
- **Gemini Q&A:** constrained prompt (practical steps, topic tag, length limits) + offline cache.
- **Gemini pack:** structured JSON checklist/tips with deterministic local fallback.

Offline capability matrix

Capability	Offline today?	Notes
Survival guides / seeded shelters	Yes	Room seeder on first launch
Pack checklists	Yes	Local cache DB
Personalized AI pack (after generate)	Yes	Stored in Room
AI Q&A (repeat questions)	Partial	Only if previously cached
I'm Safe	Yes (queue)	Flushes when online
Mesh Notes	Yes	Nearby hop + later cloud sync
Community chat	No	Requires Supabase
Map basemap tiles	No	CDN tiles; offline archives planned
OTP / verification	No	Requires Supabase (+ Twilio for SMS)

This design is effective because it matches disaster physics: **connectivity is intermittent, trust must be local, and guidance must survive without a round-trip to the cloud.**

06 What Makes This App Different?

Comparison is against common patterns in consumer disaster / safety apps—not against any single vendor claim.

Dimension	Common apps	SafePath (implemented)
Reliability	Cloud-first; blank screens offline	Room-seeded survival, packs, shelters; queued actions
Family safety	Generic location sharing always-on	Verified Family Circle + intentional I'm Safe broadcast
Community	Social feeds needing data	Online chat + offline Mesh Notes with hop/TTL controls
AI	Chatbots that die offline	Gemini with Room cache / one-shot pack persistence
Local relevance	Global generic content	Sri Lanka districts, seeded shelters (Ratnapura, Kegalle, Kalutara)
Citizen reporting	Unstructured posts	Typed severity + district + GPS alert records
Cost to user	Often ad-heavy or paywalled core	Core survival tools usable without paid unlock (see Free/Pro)
Architecture honesty	Hidden online dependency	README + code clearly mark map-tile online dependency

Competitive advantage that is real: SafePath is not 'another map.' It is a **resilience stack**—offline content + verified family signaling + peer mesh + cacheable AI—built for monsoon and landslide conditions in Sri Lanka.

07 Real-Life Disaster Scenarios

*Scenarios illustrate how **currently implemented** capabilities would change decisions. No unverified casualty-reduction percentages are claimed.*

Flood

A family in Kalutara sees rising water at night. With SafePath they open the offline pack checklist, ask AI (if previously cached or briefly online) for flood steps, check the home silhouette for nearby seeded shelters, report a HIGH flood alert with GPS, and tap I'm Safe so a verified contact knows they evacuated. Neighbors without data post a Mesh Note: 'Bridge blocked — use upper road.' Responders later see structured alerts on the map when connectivity allows. Delays removed: searching for paper checklists, unanswered calls, and unstructured WhatsApp rumors.

Landslide

In Ratnapura hill country, an advisory becomes urgent. The imminent-alert path pushes users toward the evacuation map. Seeded shelters (schools, hospital, secretariat) appear as known points. Mesh Notes carry 'slope crack on estate road' across Bluetooth hops when towers fail. Families use I'm Safe after reaching higher ground—queued if offline—so relatives stop risky return trips.

Tsunami

Coastal users need fast, calm instructions. Survival library and AI Q&A topics include tsunami guidance patterns. Community chat (when online) coordinates vertical evacuation messaging; Mesh Notes can still move 'all clear / stay high' status peer-to-peer. Reporting captures citizen observations with severity for later official correlation—not as a replacement for national warning systems.

Earthquake

After shaking, users follow offline first-aid / survival content, mark pack items used, and broadcast I'm Safe. Mesh Notes share 'building unsafe' and 'need water' without requiring a working cellular uplink. Map markers help orient to shelters once the device can load tiles or when users hand off to Google Maps navigation.

Cyclone

Preparedness dominates: AI-personalized packs emphasize shelter-in-place realities (the seeded guidance model includes cyclone severities). Checklists and tips remain available offline during multi-day outages. After the storm, queued I'm Safe messages and mesh 'road clear' notes accelerate reunion and local recovery coordination.

Urban fire

Citizens report FIRE with district + GPS. Nearby users see the alert on the map when online. Mesh Notes warn of smoke-blocked alleys. Family Circle limits panic calling by providing a deliberate safety signal once members reach a muster point.

Missing family members

Verification-gated Family Circle creates a small trusted graph. I'm Safe with coordinates (or last known / fallback) gives relatives a last intentional location ping. This does not replace official missing-person systems, but it reduces uncertainty loops that drive dangerous searching.

Medical emergency during disaster

Offline first-aid Q&A cache and survival guides support immediate care. Mesh 'need' notes can signal medical requirements to nearby devices. Shelter list includes district hospitals among seeded sites (e.g., Ratnapura / Kegalle / Kalutara hospitals)—helping civilians aim toward known facilities while professional EMS remains authoritative.

08 Human Impact

Stakeholder	How SafePath helps (based on shipped features)
Families	I’m Safe + limited trusted contacts; shared situational awareness
Elderly	Simple checklists, large alert UI, offline guides without app hopping
Children / caregivers	Pack personalization accounts for kids; clearer go-bag readiness
Tourists	District-aware map/shelters and English guidance without local knowledge
Schools	Seeded school shelters; classroom preparedness checklists
Hospitals	Visible as shelter points; community need notes can surface demand
Volunteers / NGOs	Structured citizen alerts + mesh ground truth when online sync returns
Government / DMC partners	Potential channel for verified reporting & shelter awareness (integration = future)
Responders	Map of alerts/shelters; less time decoding unstructured social noise
Local communities	Chat when online; Mesh Notes when offline—local agency preserved

Impact is measured in **better decisions under uncertainty**: fewer unanswered ‘are you safe?’ loops, faster access to checklists and shelters, and community information that can move without a tower.

09 Future Vision

Clearly labeled as future possibilities — not present claims.

Horizon	Proposal	Builds on today’s code
Near-term	Offline OsmDroid tile packs; wire Dijkstra into map UI; real push notifications	assets/tiles placeholder; RoutingEngine; notification channel
Near-term	Sinhala / Tamil UI + completed guidance translations	stepTextSi/Ta columns already on GuidanceStep
Mid-term	AI hazard prediction & personalized evacuation timing	Gemini services + alert history models
Mid-term	Organization dashboards for NGOs / DS offices	Supabase alerts, shelters, mesh sync data
Long-term	IoT rain/soil sensors, wearables, drone imagery feeds	Alert ingestion pipeline patterns
Long-term	Cross-border disaster collaboration APIs	Repository + API boundary already separated

Principle: Expand capability without abandoning the offline core. Every future module should degrade gracefully when the network disappears.

10 Free Version

*Based on current product logic (including **FamilyCircleLimits**: free = 1 contact).*

- Core offline survival library and emergency pack checklists
- Home dashboard, alert viewing, report alert, evacuation map (online tiles)
- Ask SafePath AI and AI pack personalization when API keys/network allow (cached thereafter)
- Community chat (online) and Mesh Notes (P2P)
- Family Circle with **1** trusted contact + I'm Safe
- Guest continue for exploration (local anonymous session—not full cloud identity)

Intended users: households, students, volunteers, and anyone who needs a free resilience toolkit. **Limitation:** smaller family graph; organization-scale tooling not included; map basemap still network-dependent.

11 Pro Version (Proposed)

Proposed premium model — partially foreshadowed by tier fields and contact limits in code.

Pro capability (proposed)	Why organizations / families would pay
Family Circle up to 5 contacts (limit already coded)	Multi-member households & caregivers
Priority / richer alert digests	Faster triage for dense family groups
Advanced AI insights & multi-pack profiles	Schools, estates, small clinics
Organization dashboard & exportable analytics	NGOs, DS offices, campus safety
Premium offline map packs	Field teams in dead zones
Team management & role-based access	Volunteer networks
SLA-backed support & training	Government / enterprise adoption

Pro funds sustainable maintenance of offline content, shelter verification, and AI costs while keeping life-critical free basics available to the public.

12 Business Model (Proposed)

Revenue options consistent with a public-interest disaster platform.

Stream	Description	Fit
Consumer Pro	Subscription for expanded Family Circle & AI	Households
Government licensing	District / national deployment + training	DMC, local authorities
NGO partnerships	Field packs, offline maps, reporting seats	Humanitarian orgs
Enterprise / campus	Safety dashboards for estates, universities, factories	Private sector
API access	Alert/shelter feeds for partners	Integrators
White-label	Branded resilience apps for regions	Intl. expansion
Training & support	Drills, content localization, ops support	All institutional buyers

Ethics note: Core offline survival guidance and emergency reporting should remain accessible on the free tier. Monetization targets capacity, analytics, and organizational scale—not ransoming basic safety instructions.

13 Social Impact

- **Protected decision-making:** households act on checklists, shelters, and calm AI guidance instead of rumor.
- **Time saved:** I'm Safe and mesh notes reduce repeated calling and searching.
- **Communication resilience:** P2P notes move when towers do not.
- **Community agency:** citizens contribute structured alerts and local hazard context.
- **Reduced panic loops:** verified family signals replace uncertainty spirals.
- **Preparedness culture:** personalized packs make monsoon readiness tangible before peak season.
- **Coordination substrate:** shared map + alert data improves NGO/responder common operating pictures when synced.

SafePath does not claim to prevent disasters. It claims to **improve the human system around disasters**—information, family trust, and local coordination—using technology that still works when infrastructure does not.

14 Why This Platform Matters

SafePath matters because it advances disaster technology from **reactive broadcasting** toward **resilient, two-way, offline-capable community infrastructure**.

- Faster access to critical instructions without waiting for a working data session for every tap.
- Better coordination signals among families and neighbors via I’m Safe and Mesh Notes.
- Empowerment of communities to report and share ground truth in structured form.
- Support for families as first responders to each other—the real first mile of every crisis.
- A path from reactive scrolling to proactive preparedness (packs, drills, cached AI).
- A credible technical foundation (Room, WorkManager, Nearby, Supabase, Gemini) that institutions can extend—not a demo-only prototype narrative.

15 Closing Vision

Imagine a Sri Lanka where every household carries a phone that still knows what to do when the signal bars disappear: a checked go-bag, a known shelter, a calm next step, a way to tell family ‘I’m safe,’ and a neighborhood note that hops from device to device until help can hear it.

That future is not science fiction in this repository. The offline core, verification-gated family safety, mesh transport, citizen alerting, and cacheable AI assistance are already implemented. What remains—deeper offline maps, fully wired on-device routing, richer notifications, and institutional dashboards—is an execution roadmap, not a leap of faith.

SafePath’s promise is simple and serious: **make preparedness personal, make safety signals trustworthy, and make coordination possible even when the network is not.**

SafePath — Survival tools that work when the network does not.

Team Hades · July 11, 2026
github.com/godragun/SafePath

A Appendix — Implementation Status Matrix

Area	Status	Evidence
Email auth / guest	Working	AuthViewModel
Phone/email OTP	Working	SupabaseVerificationService
Family Circle + I’m Safe	Working	Family + SafeBroadcast + Worker
Emergency pack UI	Working	PackRepository / checkboxes
AI pack personalize	Working	GeminiPrepService
Ask SafePath AI	Working	GeminiQaService + Room cache
Survival library	Working	SurvivalRepository
Report alert	Working	AlertRepository → Supabase
Evacuation map	Partial	Leaflet online tiles; Google Maps nav
Dijkstra routing	Library only	Not wired to map UI
Mesh Notes P2P	Working	Nearby + Room + MeshSyncWorker
Community chat	Working	Supabase realtime
Push notifications	Partial	Channel + preference only
Offline map tiles	Planned	assets/tiles placeholder
Si/Ta localization	Partial	DB columns; placeholder text
Situm SDK	Not present	No dependency in project

Document control

Generated from codebase analysis of SafePath (com.hades.safepath). Date: July 11, 2026. This document is intended for competition judges, investors, and institutional partners. All ‘Working/Partial/Planned’ labels reflect repository state at generation time.